ITCS216: Fundamentals of Database Systems (1/2025)

Faculty of Information and Communication Technology

In-class Exercise 06: Mapping EERD to Relational Schema

Objectives:

- Transform the given Entity Relationship Diagram to a relational model step by step

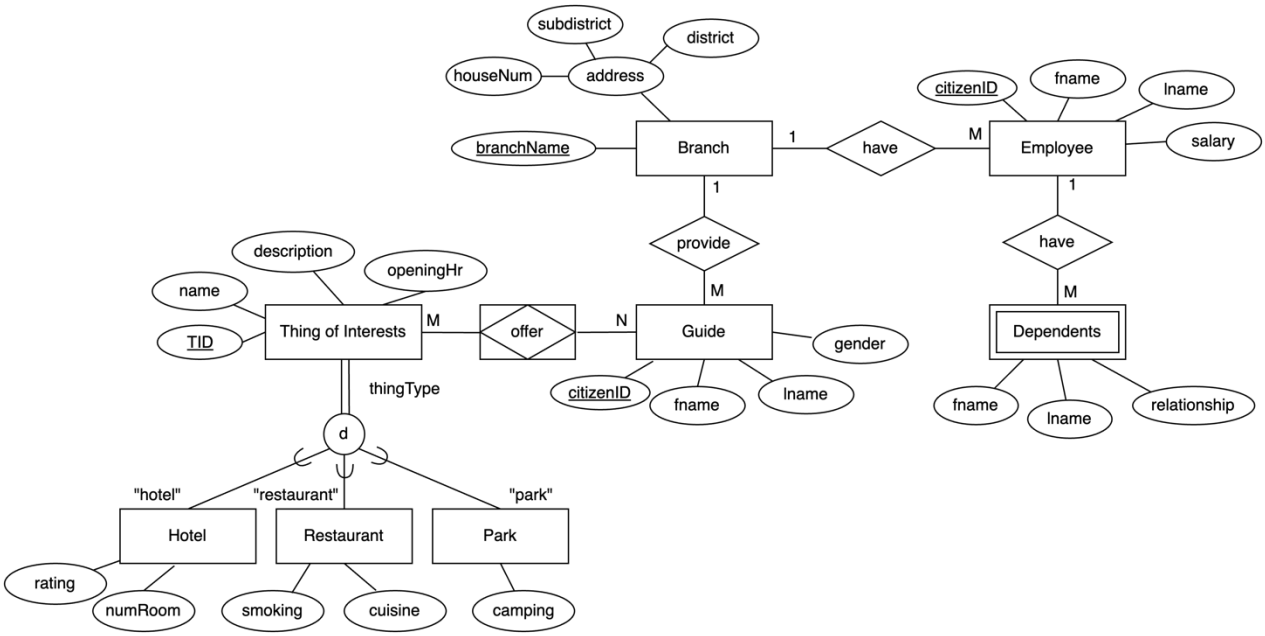
Instructions:

1. From the given ER diagram, use 8-step transformation to create a relational model.
   Note: you must show the step-by-step transformation
2. Draw the final relational model

Given the following EER diagram, use 8-step transformation to create a relational model, including

1) Transform regular entities including simple and composite attributes
2) Transform weak entities
3) Transform 1:1 relationship
4) Transform 1:M relationship
5) Transform M:M relationship
6) Transform multivalued attributes
7) Transform N-ary relationship
8) Transform specialization or generalization

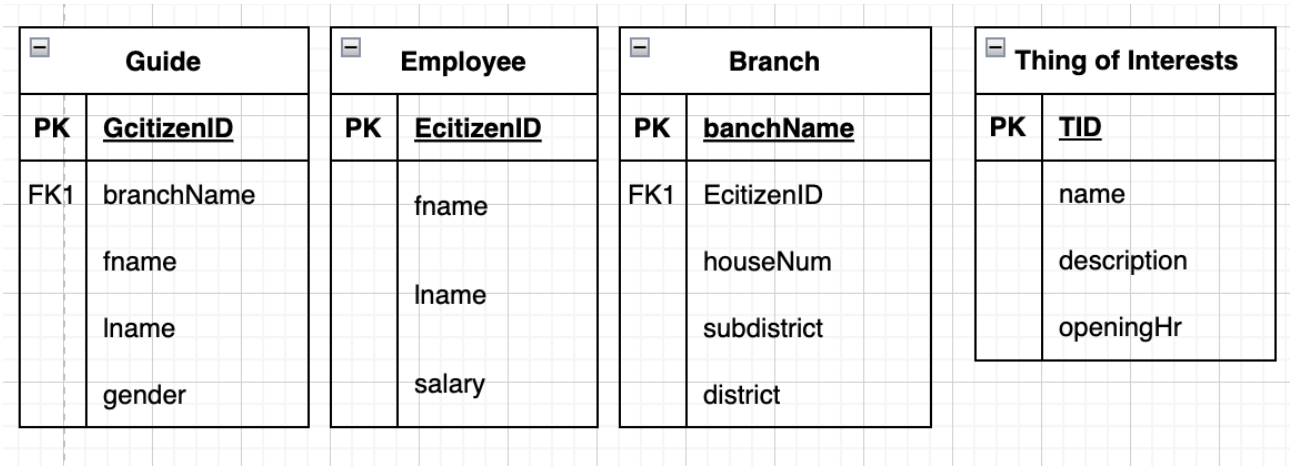
Note: If some steps are *not applicable* in the ERD, i.e., no that entities, attributes, or relationship, etc., please clearly specify "**Not applicable**" in the step.



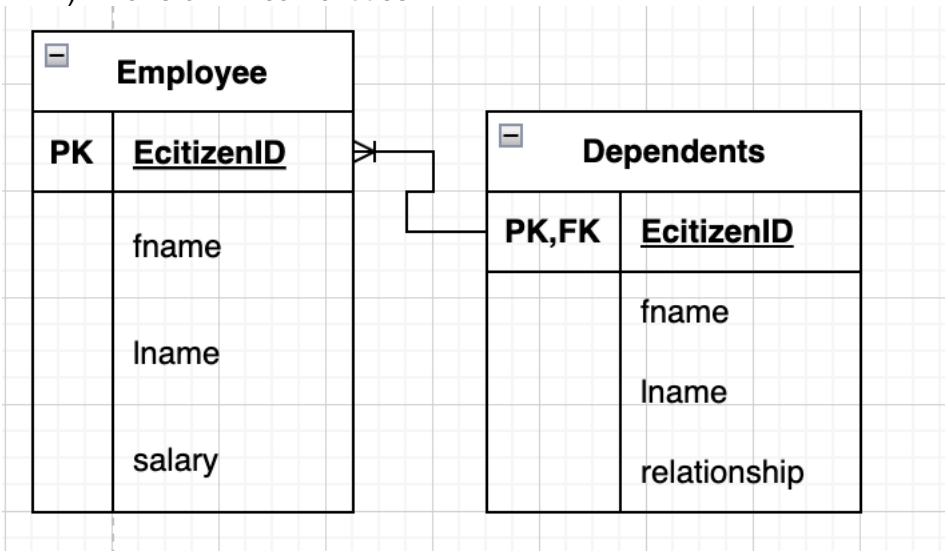
Before starting to draw the final relational model, please complete this table below

Step	**Found in EERD? (Yes/No)**	**Which entities are involved? What changes?**
1 Regular	Yes	Employee, Dependents, Branch, Guide, Thing of Interests, Hotel, Restaurant, and Park. The attribute of Branch, address got excluded.
2 Weak	Yes	Dependents The citizenID from Employee assigned to Dependents
3 1:1	Yes	Total of Interests All of the subclass of Total of Interest will share the same FK from its superclass
4 1:M	Yes	Branch get the EcitizenID from Employee Guid get the branchName from Branch Dependents get FK from Employee
5 M:N	Yes	Offer get TID as PK and FK Offer get GcitizenID as PK and FK
6 Multivalued	No	
7 N-ary	No	
8 Specialization	Yes	Total of Interests has three subclass, and both of them share the same PK from its superclass.

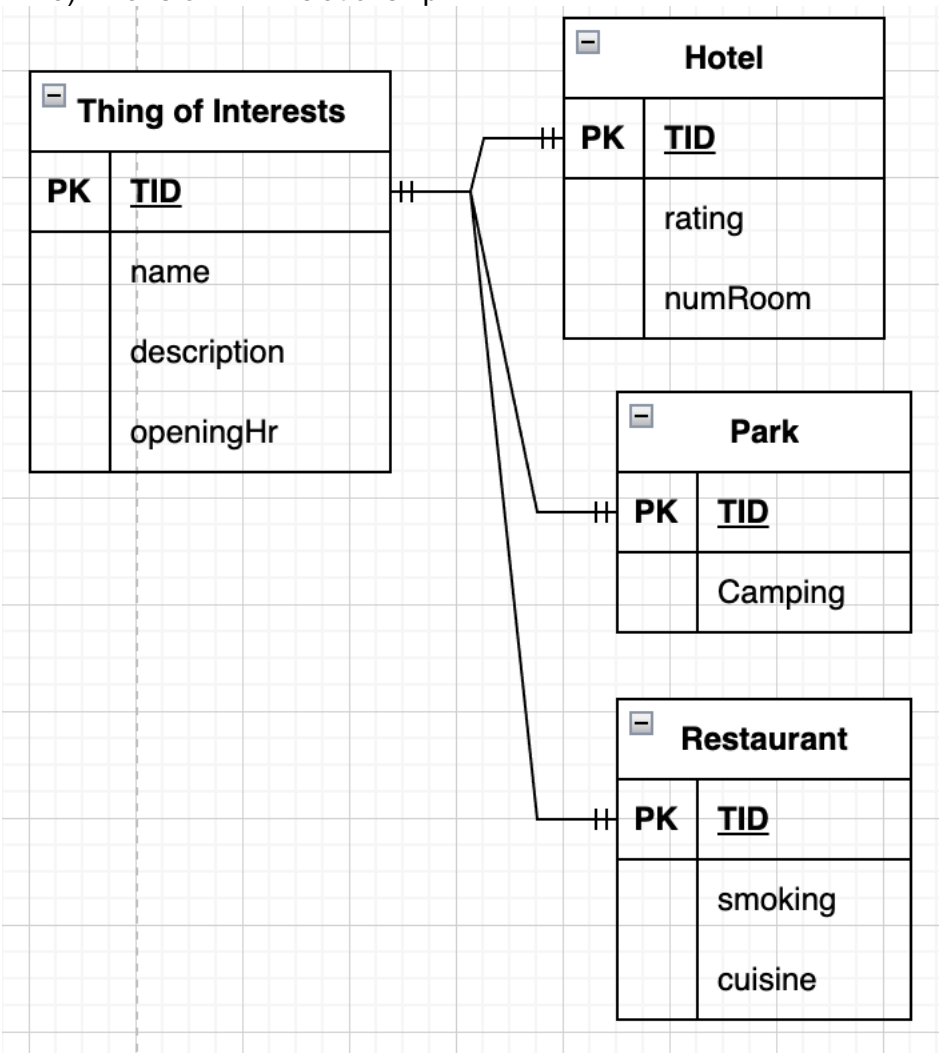
1) Transform regular entities including simple and composite attributes



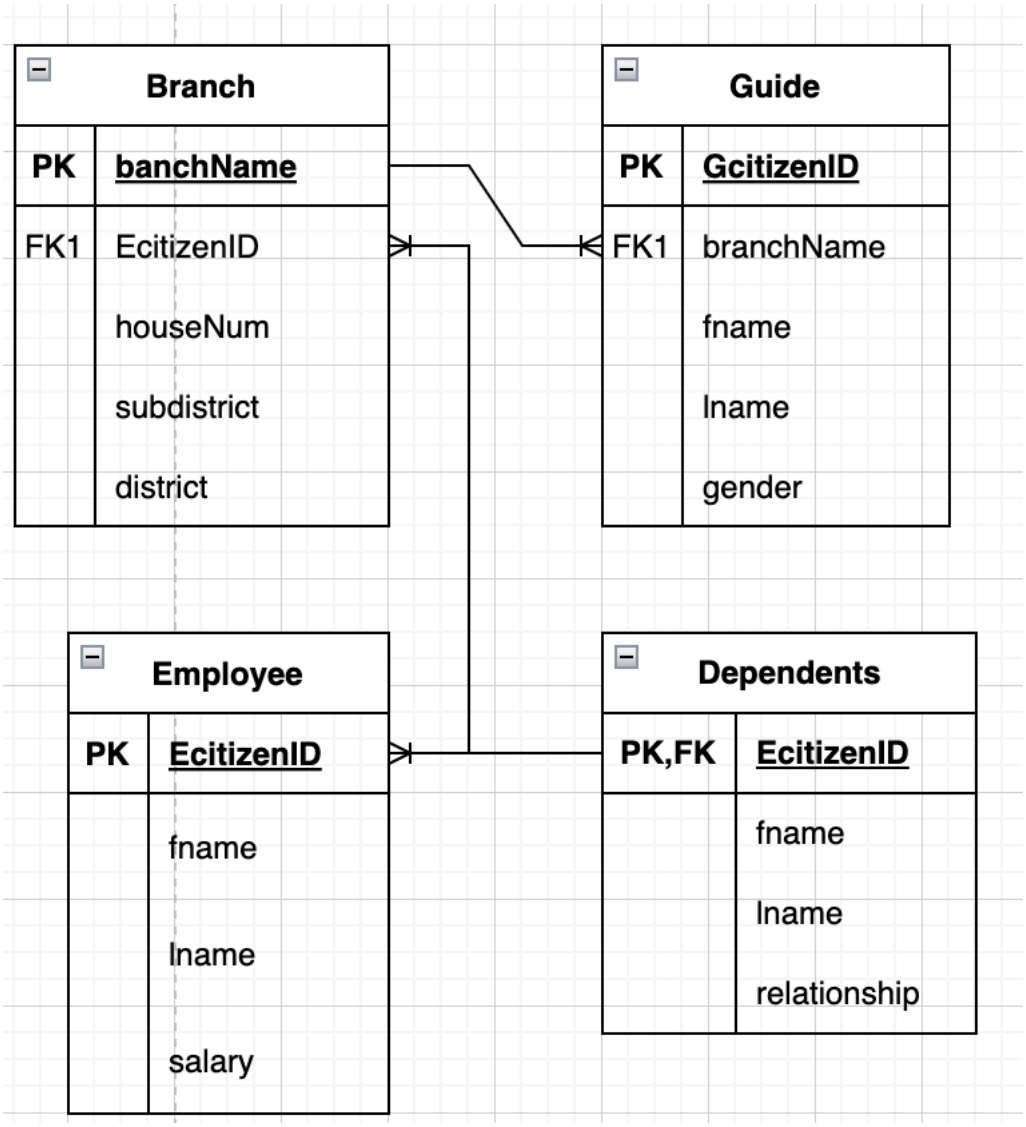
2) Transform weak entities



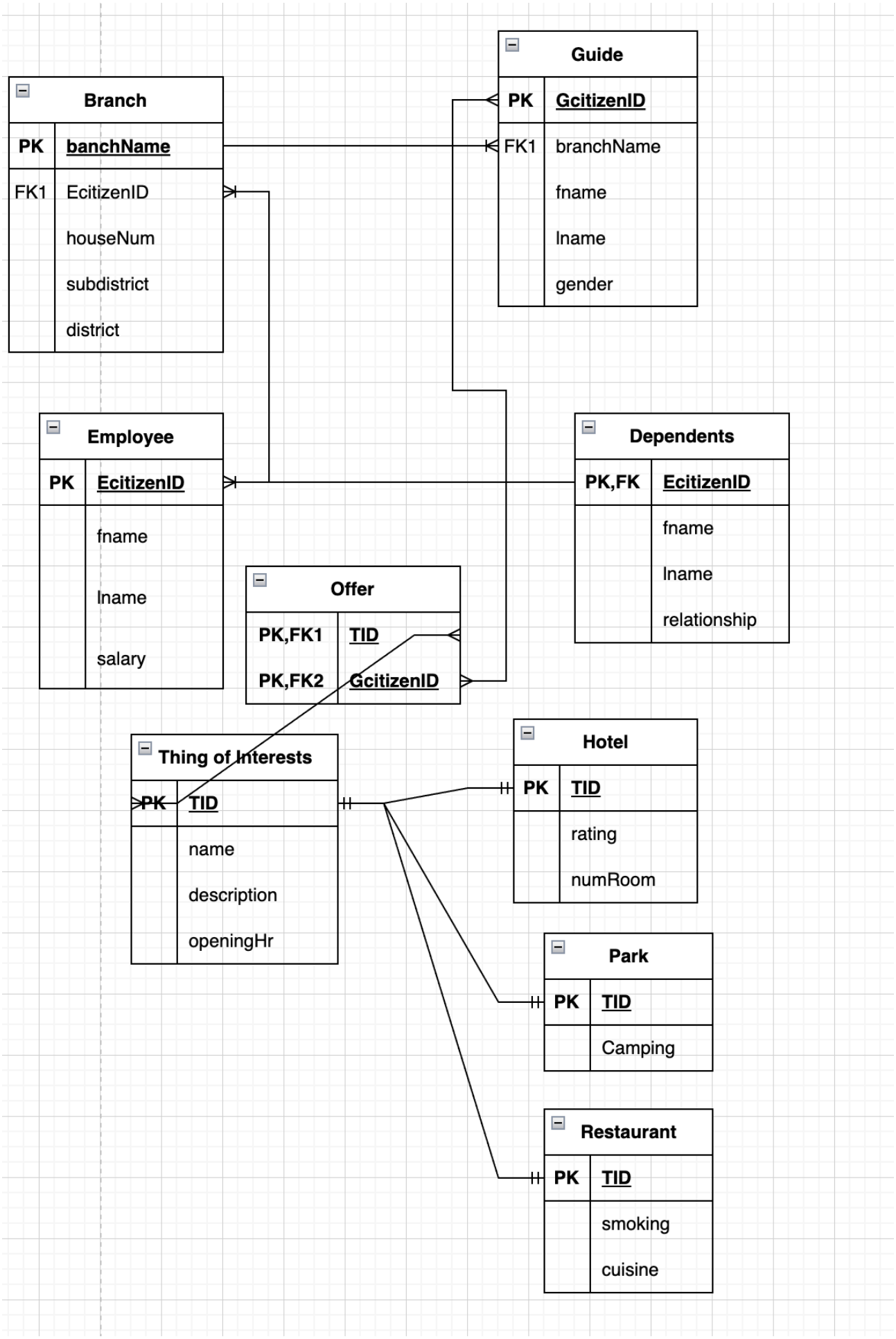
3) Transform 1:1 relationship



4) Transform 1:M relationship


Branch
PK
banchName
FK1
EcitizenID
houseNum
subdistrict
district
Guide
PK
GcitizenID
FK1
branchName
fname
lname
gender
Employee
PK
EcitizenID
fname
lname
salary
Dependents
PK,FK
EcitizenID
fname
lname
relationship

5) Transform M:M relationship


Guide
PK GcitizenID
FK1 branchName
fname
lname
gender
Branch
PK banchName
FK1 EcitizenID
houseNum
subdistrict
district
Employee
PK EcitizenID
fname
lname
salary
Dependents
PK,FK EcitizenID
fname
lname
relationship
Offer
PK,FK1 TID
PK,FK2 GcitizenID
Thing of Interests
PK TID
name
description
openingHr
Hotel
PK TID
rating
numRoom
Park
PK TID
Camping
Restaurant
PK TID
smoking
cuisine

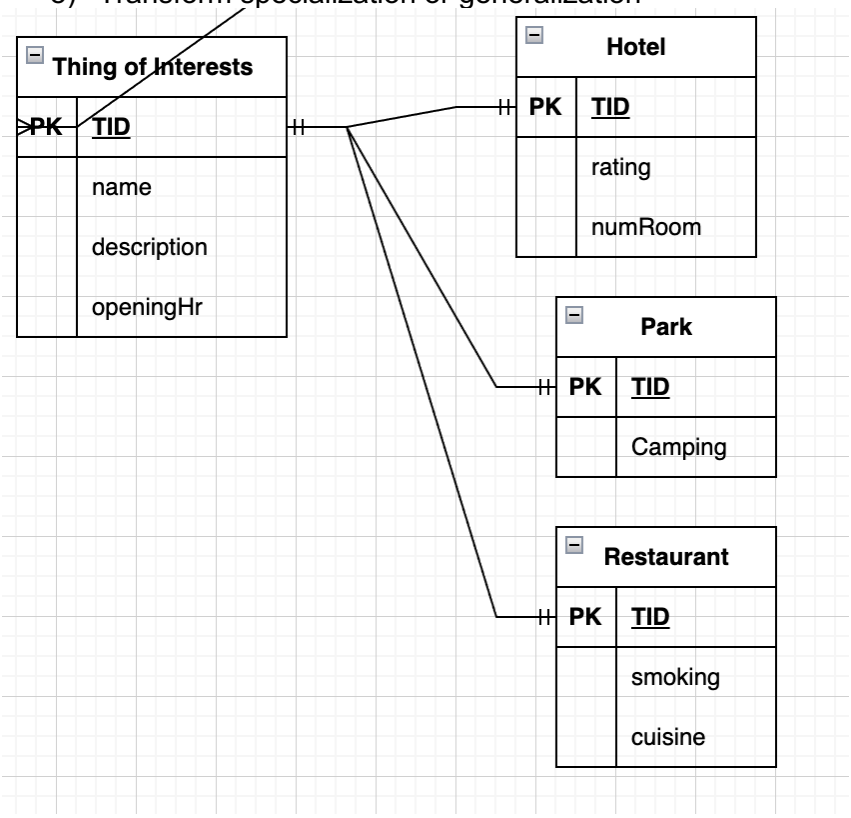
6) Transform multivalued attributes

No multivalued attributes in the graph

7) Transform N-ary relationship

No N-ary relationship in the graph

8) Transform specialization or generalization



Final Relational Model

